

[Secure Exit](#)SIMATS
ENGINEERING

SIMATS Online Programming Lab

JAVA PROGRAMMING

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CO3-AT3-Assignment

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QUESTIONS

Q1Banking Transaction System –
Exception Handling
10 marks**Q2**Hospital Patient Registration –
User-Defined Exception
10 marks**Q3**E-Commerce Order Processing
– Built-in Exceptions
10 marks**Q4**Hospital Emergency System –
Thread Priority
10 marks**Q5**Online Food Delivery – Multiple
Threads and Execution Order
10 marks

5/5 answered

Q1

Banking Transaction System – Exception Handling

10
marks

Develop a Java program for an ATM withdrawal system. The program accepts the account balance and withdrawal amount.

The program must:

- Accept the current account balance and withdrawal amount.
- Throw a user-defined `InsufficientBalanceException` when the withdrawal amount exceeds the balance.
- Handle withdrawal amounts that are zero or negative.
- Handle invalid numeric input using a suitable built-in exception.
- Display a meaningful transaction result.
- Use try-catch-finally so that the transaction completion message is always displayed.

Sample Test Cases:

Input / Scenario	Expected Result	Purpose
10000 3000	Normal case	
10000 12000	Boundary / exception case	
10000 -500	Invalid input	
abc 3000	Invalid input	

Your Code

```
1 import java.util.InputMismatchException;
2 import java.util.Scanner;
3 class InsufficientBalanceException extends E
4     public InsufficientBalanceException(Stri
5         super(message);
6     }
7     public static void main(String[] args)
8     {
9         Scanner scanner = new Scanner(System
10     try
11     {
12         double balance = scanner.nextDouble
13         double withdrawal = scanner.nextDou
14         if(withdrawal < 0)
15         {
16             throw new IllegalArgumentExcept
17         }
18         if(withdrawal > balance)
19         {
20             throw new InsufficientBalanceEx
21         }
22         System.out.println("normal case"
23     }
24     catch(InputMismatchException | Illeg
25     {
26         System.out.println(e.getMessage(
27     }
28     catch(InsufficientBalanceException e
29     {
30         System.out.println(e.getMessage(
31     }
32     }
33 }
```



Input

1000
500



Output

normal case

✓ Submitted

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